

ECE240:PCS

Problem Set #2

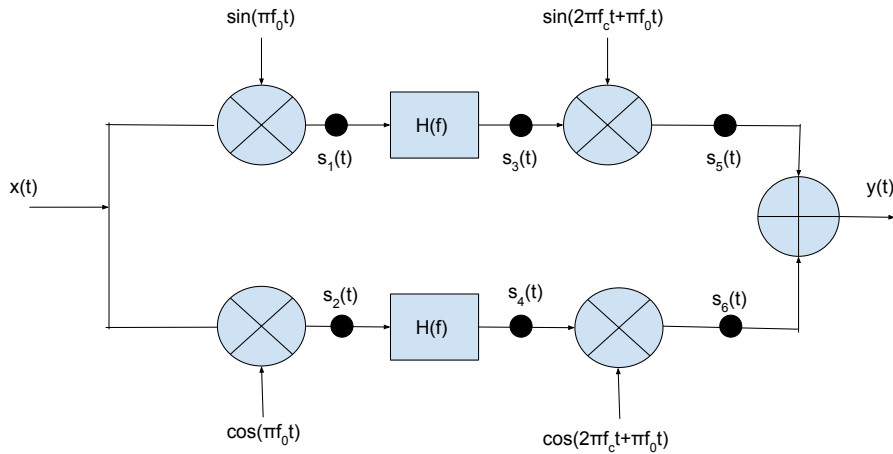
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Problem 1 Properties of Hilbert Transform:

- Show that Hilbert transform of an even signal is odd and an odd signal is even.
- Show that a signal $x(t)$ and its Hilbert transform are orthogonal.
- Show that energy content of a signal and its Hilbert transform are the same.
- Show that Hilbert transform of the derivative of a signal is same as the derivative of the Hilbert transform of the signal.
- Let $x(t)$ represent a bandpass signal and $m(t)$ denote a lowpass signal with non-overlapping spectra. Show that Hilbert transform of $c(t) = x(t)m(t)$ is given by $x(t)\hat{m}(t)$.

Problem 2 Consider the following block-diagram. Let $X(f)$ be a triangular frequency response from $-f$ to f with magnitude 1, and $H(f)$ be a rectangular response between $-f/2$ and $f/2$ with magnitude 1. Answer the following:



1. Sketch the Fourier transform of $s_1(t)$ and $s_2(t)$.
2. Sketch the Fourier transform of $s_3(t)$ and $s_4(t)$.
3. Sketch the Fourier transform of $s_5(t)$ and $s_6(t)$.

4. Sketch the Fourier transform of $y(t)$.
5. Fill in the blanks by comparing $X(f)$ and $Y(f)$: $y(t)$ is a _____ modulated signal of $x(t)$.
6. Can a Hilbert transformer block be used to reduce the number of local oscillators in the block diagram above. Carefully point out the possible location of such a block (by rewriting some parts of the block diagram).

Problem 3 Let $x(t) = m(t)\cos 2\pi f_c t + \hat{m}(t)\sin 2\pi f_c t$. Write the complex baseband representation of $x(t)$. Plot the Fourier transform of $x(t)$. What is its bandwidth.

Problem 4 Learn the construction of the Weaver's modulator and demodulator.

Problem 4 The message signal $m(t)$ with peak frequency deviation given by $25\text{Hz}/V$ is shown in the figure. Plot the frequency deviation in Hz and phase deviation in radians.

Problem 6 A message signal $m(t)$ has a bandwidth of 10KHz and a peak amplitude of $1V$. Estimate the bandwidth of the frequency modulated wave when the signal modulates a carrier with peak frequency deviation of (i) $10\text{Hz}/V$ (ii) $100\text{Hz}/V$ (iii) $1000\text{Hz}/V$.

Problem 7 (Sampling) Problem 6.1-1 - 6.1-5, 6.1-8 from reference book 2.

Problem 8 (Quantization) Problem 6.2-1 to 6.2-4, 6.2-6, 6.2-8 - 6.2-10 from reference book 2.

Problem 9 Solved examples in Leon Garcia, book uploaded.

Problem 10 Please be ready for 1 mark questions from the rest of the syllabus on line codes, Nyquist criterion for zero-ISI, matched filtering and entropy.